SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: Total Marks: 50 Annexure A

Code of Conduct:

I ***_P.Sunayana(192424303)_*** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: 

CSA5112 - Cryptography And Network Security

P. Sunayana.
192414303.

CO2 - Assessment 1 (Analytical Problem Solving)

1) RSA Encryption Analysis :-

Given; $p = 11$

$q = 13$

public exponent $(e) = 7$.

Message $(M) = 9$.

a) Compute n and $\phi(n)$:-

Calculate the modulus :- $n = p \times q$

$n = 11 \times 13 = 143$.

Calculate the Euler's totient function :-

$$\phi(n) = (p-1)(q-1)$$

$$\phi(n) = (11-1)(13-1)$$

$$\phi(n) = 10 \times 12 = 120.$$

b) Choose $e = 7$ and find private key d :-

$$\gcd(e, \phi(n)) = 1$$

Since $\gcd(7, 120) = 1$, the selected value is valid.

The private key exponent d :-

$$ed \equiv 1 \pmod{\phi(n)}$$

$$7d \equiv 1 \pmod{120}$$

let $d = 103$;

$$7 \times 103 = 721 = 6(120) + 1$$

$$d = 103$$

Keys :- public key :- $(e, n) = (7, 143)$

private key :- $(d, n) = (103, 143)$.

c) Encrypt the message :- (M=9).

RSA encryption formula:-

$$C = M^e \bmod n$$

$$C = 9^7 \bmod 143$$

(using modular exponential)

$$9^2 = 81$$

$$9^4 = 81^2 = 6561 = 126 \pmod{143}$$

$$C = (126 \times 81 \times 9) \bmod 143$$

Since $126 \times 81 \equiv -1 \pmod{143}$

$$C = (-9) \bmod 143 = 134.$$

Ciphertext :-

$$C = 134.$$

d) Decrypt the ciphertext and verify the result :-

RSA decryption formula:-

$$M = c^d \bmod n$$

$$M = 134^{103} \bmod 143.$$

$$M = 9.$$

Verification:- The decrypted message matches the original one.

e) Analyze how changing e affect security :-

→ The public exponent e is used for encryption and signature verify.

→ It must be relatively prime to $\phi(n)$.

→ the small exponent can improve performance.

→ the large exponent generally increases the cost exponentiation.

Conclusion:- RSA successfully encrypts & decrypts message. the small values of P & Q make this example insecure in Practice. Real RSA requires sufficiently large keys & secure padding.

2) Block Cipher Mode Evaluation:-

Given;

$P = [1010, 1010, 1111, 1010]$.

$k = 1100$, $IV = 0000$.

Assume a simplified 4-bit block cipher:-

$$E_k(X) = X \oplus k$$

where; $\oplus$ indicates bitwise XOR.

a) Encrypt using ECB and CBC modes:-

1) ECB Mode:-

$$C_i = E_k(P_i)$$

$$C_i = P_i \oplus k_i \quad \text{(assumed cipher)}.$$

Block	Plaintext (P_i)	key (k)	Ciphertext (C_i)
1.	1010	1100	0110
2.	1010	1100	0110
3.	1111	1100	0011
4.	1010	1100	0110

$\therefore C_{ECB} = [0110, 0110, 0011, 0110]$.

2) CBC mode:-

$$C_i = E_k(P_i \oplus C_{i-1})$$

$C_0 = IV = 0000$ (for first Block).

Block 1:- $P_1 \oplus IV = 1010 \oplus 0000 = 1010$

$C_1 = 1010 \oplus 1100 = 0110$

Block 2:- $P_2 \oplus C_1 = 1010 \oplus 0110 = 1100$

$C_2 = 1100 \oplus 1100 = 0000$

Block 3:- $P_3 \oplus C_2 = 1111 \oplus 0000 = 1111$

$C_3 = 1111 \oplus 1100 = 0011$.

Block 4:- $P_4 \oplus C_3 = 1010 \oplus 0011 = 1001$.

$C_4 = 1001 \oplus 1100 = 0101$.

$\therefore C_{CBC} = [0110, 0000, 0011, 0101]$.

b) Compare the ciphertext outputs:-

Feature	ECB Mode	CBC Mode
Encryption method	Encrypts each block independently	Combines each plaintext with previous block
IV requirement	Not required	Required.
Identical plaintext blocks	Produce identical ciphertext blocks.	Can produce different ciphertext blocks.
Error propagation	Errors generally affect the corresponding block.	Errors can affect the current blocks.
Pattern hiding	Poor.	Better than ECB when used correctly.

c) Analyze which mode hides patterns better:-

→ In ECB, identical plaintext blocks produce identical ciphertext blocks. In this example, plaintext blocks 1, 3 and 4 are all 1010, so the ciphertext blocks are 0110.

→ In CBC, each block depends on the previous ciphertext block. Therefore, repeated plaintext blocks can produce different ciphertext blocks.

3) DES (vs) 3-DES (vs) Blowfish performance study:-

Given;

DES – 56 bits (size) - 1 ms.

3-DES – 168 bits (size) - 6ms.

Blowfish – 128 bits (size) – 1ms.

Total data blocks = 1000.

(i) Calculate the total encryption time for each algorithms:-

Formula:-

Total Time = No. of Blocks * Time per Block

1) DES:-

$T = 1000 \times 1$

$T = 1000\ ms = 1$ seconds. (Moderate)

2) 3-DES:-

$T = 1000 \times 6$

$T = 6000\ ms = 6$ seconds. (Slowest)

3) Blow-Fish:-

$T = 1000 \times 1$

$T = 1000\ ms = 1$ seconds.

(ii) Analyze the trade-off's between security and performance:-

1) DES:-

→ Uses a 56-bit Effective key.

→ Encryption is faster than 3-DES under the timings.

→ Its short key makes it vulnerable to brute-force attacks.

→ It is not suitable for protecting modern sensitive information.

(ii) 3 - DES :-

→ Applies the DES algorithm three times.

→ provides stronger than single DES by the 6 - second result.

→ Requires more processing time, as shown by the 6 - second result.

→ It is now considered a legacy algorithm and is unsuitable for most new systems.

(iii) Blow - Fish :-

→ Supports variable - length keys, including 128 - bit keys.

→ Has the shortest encryption time in the given data.

→ Offers (11) better key - length options than DES.

→ Its 64 - block size creates limitations when encryption large volumns of data.

(iii) Recommend the best algorithm for secure real - time communication :-

→ Among all three algorithms, Blowfish is the best choice based on the provided performance figures, because it completes encryption in only 1 second, for 1000 blocks and supports a 128 - bit key.

→ However, this recommendation is limited to the supplied choices; Blowfish's 64 - bit block size is a concern for modern applications, while DES and 3 - DES are legacy algorithms.

4) Diffie-Hellman key Exchange Analysis:-

Given;

$p = 23$.

$g = 5$

$a = 6$.

$b = 15$.

a) Public key of user A:-

$$Y_A = g^a \bmod p$$

$$Y_A = 5^6 \bmod 23 = 8.$$

b) public key of user B:-

$$Y_B = g^b \bmod p.$$

$$Y_B = 5^{15} \bmod 23 = 19.$$

c) Shared secret key:-

User A:-

$$k_A = 19^6 \bmod 23$$

$$k_A = 2.$$

User B:-

$$k_B = 8^{15} \bmod 23$$

$$k_B = 2.$$

$$k_A = k_B = 2.$$

Therefore; the shared secret key is 2.

d) Demonstrate a Man - in - the - Middle (MITM) attack:-

→ A man - in - the middle attacker intercepts and modifies the publickey exchange between A and B.

For illustration; Suppose the attacker uses a private key $x = 3$.

The attacker's public value is :-

$$Y_E = 5^3 \bmod 23.$$

$$Y_E = 125 \bmod 23 = 10.$$

If A recieves the attacker's public value 10, A calculates:-

$$k_{AE} = 10^6 \bmod 23.$$

$$k_{AE} = 6$$

For B; $k_{BE} = 10^{15} \bmod 23 = 19.$

Thus, A and B may establish different keys with the attacker.

e) Prevention :-

→ use digital signatures to authenticate public keys.

→ Use digital certificates from a trusted certificate authority.

→ Use authenticated key-exchange protocals such as TLS.

Conclusion:- Diffie-Hellman establishes a shared secret, but authentication is necessary to prevent MITM attacks.

5) ECC (vs) RSA comparsion:-

Given;

RSA key size = 1024 bits.

Ecc key size = 160 bits.

Device : Resource-limited Iot devices.

a) Computational Cost:-

RSA :- uses large - integer modular exponentiation and may require significant processing.

ECC:- uses elliptic - curve (authenticate) arthmetic and achieves comparable security with smaller keys under the question's assumptions.

b) power and Memory consumption:-

Feature	RSA	ECC
key size	larger.	Smaller.
Memory usage	Higher.	lower.
Communication overhead	Higher.	lower.
Powa consumption	Generally Higher.	Generally lower in efficient implementation

c) Best Cryptosystem for Iot :-

ECC is preferrable because:-

1) It (is) uses smaller keys.
2) It requires less memory for key storage.
3) It reduces communicated overhead.
4) It can reduce processing and energy requirements.

Conclusion:- ECC is better suited to the given Iot scenario. However, the specified keys sizes are too small for modern deployments; secure, currently recommended parameter should be used.